

## **Backtesting VaR Models: The Case of Commodities**

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### **Abstract**

One of the most widely used methods to quantify risk is ‘Value at Risk’. VaR models are useful only if they predict future risks accurately. This paper focuses on a comparative evaluation of three broad approaches to calculate VaR for a total of nine commodities traded on Multi Commodity Exchange of India. The primary objective of the study is to identify the most accurate VaR model for the commodity asset class in general. VaR is calculated using five different methods (two methods each of parametric & non-parametric approaches and one method of semi-parametric approach) for all nine commodities for a period of nine years starting October 2006 till October 2015. To identify the better performing VaR methods accurately, the analysis is performed in two phases, Pre-Crisis (October 2006 to December 2009) and Post Crisis (January 2010 to October 2015). Volatility Weighted Historical Simulation (VWHS) VaR method outperformed other methods in both parts of the analysis exhibiting a success ratio of 100% each time. We also conclude that the selection of similar or contrasting data periods in terms of market conditions for VaR calculation and VaR backtesting affects the performance of VaR methods in general. These findings are relevant for retail and institutional investors who hold commodities in their portfolios and traders who need to calculate VaR for their commodity portfolios.

**Key Words:** *Value at Risk, Backtesting, Historical VaR, Bootstrapping, Volatility Weighted Historical Simulation VaR, Normal VaR, GARCH (1,1) VaR*

**JEL Classification Codes:** C52; C53; G21

## 1. Introduction

Over the past few decades, risk management has evolved to a point where it is considered to be a distinct sub-field in the theory of finance. The growth of risk management industry traces back to the increased volatility of financial markets in 1970's. The breakdown of Bretton Woods system of fixed exchange rates and the rapid advance of new theory, such as adoption of Black-Scholes model, were among the important events that contributed to this 'risk management revolution'. Another factor is simply the fact that trading activity increased significantly (Linsmeier & Pearson, 1996; Dowd, 1998). For instance, the average number of shares traded per day grew from 3.5 million in 1970 to 40 million in 1990 (Dowd, 1998). At least equally impressive was the growth of the dollar value of outstanding derivatives positions; from \$1.1 trillion in 1986 to \$72 trillion in 1999 (Jorion, 2001). These elements combined with the unpredictable events in 1990's, such as financial disasters in Barings Bank, Orange County, Daiwa and Metallgesellschaft, highlighted the need for improved internal risk management tools (Dowd, 1998; Jorion, 2001).

Risk is an important determinant of investment decision for an asset. Measurement of risk has evolved over time since the advent of Harry Markowitz's idea of using standard deviation as a proxy for risk in 1952. Risk has been defined and measured in different ways by researchers, academicians and practitioners. One of the most widely used methods to quantify risk is 'Value At Risk' (commonly known as VaR). During the past decade, VaR has become one of the most popular risk measurement techniques in finance. VaR is a method which aims to capture the market risk of a portfolio of assets. According to Bank for International Settlements, market risk is defined as the risk of losses in on and off balance sheet positions arising from movements in market prices. Put formally, VaR measures the maximum loss in value of a portfolio over a predetermined time period for a given confidence interval.

The mathematical roots of VaR were developed already in the context of portfolio theory by Harry Markowitz and others in 1950's. Financial institutions began to construct their own risk management models in 1970's and 1980's, but it was not until the pioneering work from J.P. Morgan and their publication of Risk Metrics system in 1994 that made VaR the industry-wide standard (Dowd, 1998) and (Jorion, 2001). Risk Metrics was originally an Internet-based service

with the aim to promote VaR as a risk management method. The service provided free data for computing market risk. Later, Risk Metrics became an independent consulting and software firm. The Basel Capital Accord of 1996 played a significant role as it allowed banks to use their Internal VaR models to compute their regulatory capital requirements. (Linsmeier & Pearson, 1996) Since then, VaR has been one of the most used measures of market risk. The concept of Value at Risk (VaR) has quickly become the standard measurement of risk in the financial sector for both financial institutions and financial regulators (Engle & Manganelli, 1999) for its ease of use and clarity of meaning.

VaR was originally developed to measure market risk, which is caused by movements in the level or volatility of asset prices (Jorion, 2001). According to (Dowd, 1998), market risks can be subdivided into four classes: interest rate risks, equity price risks, exchange rate risks and commodity price risks. (Linsmeier and Pearson, 1996) give the following formal definition for VaR:

*“Using a probability of  $x$  percent and a holding period of  $t$  days, an entity’s value at risk is the loss that is expected to be exceeded with a probability of only  $x$  percent during the next  $t$ -day period.”*

The basic idea behind VaR is straightforward since it gives a simple quantitative measure of portfolio’s downside risk. VaR has two important and appealing characteristics. First, it provides a common consistent measure of risk for different positions and instrument types. Second, it takes into account the correlation between different risk factors. This property is absolutely essential whenever computing risk figures for a portfolio of more than one instrument (Dowd, 1998).

Under the 1997 Amendment to the Basel Accord and the second Capital Adequacy Directive (CAD) adopted by the European Union, banks are able to seek approval for the adoption of their own in-house VaR models in order to calculate the minimum regulatory required capital to cover their market risks. Given that banks are permitted to develop differing VaR models, it is necessary for supervisory bodies to be able to assess the relative performance of these alternative models. This issue is highlighted by (Hendricks and Hirtle, 1997, pp. 8-9):

*“The actual benefits to be derived from the VaR estimates depend crucially on the quality and accuracy of the models on which the estimates are based. To the extent that these models are inaccurate and misstate the banks’ true risk exposures, then the quality of the information derived from any public disclosure will be degraded. More important, inaccurate VaR models or models that do not produce consistent estimates over time will undercut the main benefit of a models-based capital requirement: the closer tie between capital requirements and true risk exposures. Thus, assessment of the accuracy of these models is a key concern and challenge for supervisors.”*

(Engel and Gizycki, 1999) identify three banners under which such decisions can be made. These are conservatism, accuracy and efficiency. Supervisory bodies are not only concerned with the accuracy of potential models but also the level of conservatism regarding the estimated measure of risk. A conservative model can be defined as one that produces consistently high estimates of risk relative to other models. In addition, firms are concerned with model accuracy. They wish to adopt VaR models that will satisfy capital adequacy requirements and thus be sufficiently conservative to please the supervisors. Thirdly, firms want the model to be efficient in order to minimize the level of capital reserves that must be held so that surplus reserves can be used for other purposes like lending and operations. Furthermore, it is of interest to firms to be able to evaluate the performance of contesting models prior to adoption of any one model. It will be expensive in terms of both time and money for a firm to change, once any one model has been adopted as best practice within the firm. Over recent years the range of techniques available to risk managers has increased vastly, therefore the decision of which methodology to adopt is no longer straightforward as many alternative approaches can all be considered for calculation of VaR.

VaR can be used to measure the market risk of any portfolio. It is a multi-purpose tool of managing risk and meeting regulatory requirements. Abundant research exists on identification of a suitable VaR model for equity indices, bond portfolios, foreign exchange rates etc. The idea of determining an accurate methodology for calculation of VaR has been pursued through the evaluation technique of Backtesting. Backtesting is a statistical procedure where actual profits and losses are systematically compared to corresponding VaR estimates. For example, if the

confidence level used for calculating daily VaR is 99%, we expect an exception to occur once in every 100 days on average. In the backtesting process we could statistically examine whether the frequency of exceptions over some specified time interval is in line with the selected confidence level.

The aim of this study is to identify an accurate and efficient VaR methodology for commodities. Commodities exhibit certain risk characteristics that are different from traditional assets like stocks and bonds. A good value-at-risk (VaR) model must capture those characteristics.

First, many commodities, such as agricultural products, are not storable. Others, such as livestock and energy products like electricity, can only be stored at very high costs. As a result, supply or demand changes are translated immediately into price changes, which lead to higher volatility of commodity investments compared to traditional assets.

Second, supply and demand shocks occur more frequently and on a larger scale. Drought or frost can lead to an unexpected decrease in the supply of agricultural products and a subsequent sharp increase in prices. Natural disasters can also affect commodity prices. For example, when Hurricane Katrina hit New Orleans in 2005, the damage to petroleum production and refinery capabilities led to an increase in crude oil prices. Political instability in oil-exporting countries accounts for the additional variation in oil prices. Demand shocks occur less frequently and are generally on a smaller scale, but can significantly affect commodity prices.

The third source of commodity price variation is the lack of governmental control. While central banks can influence stock and bond markets they cannot compensate for supply shocks or changes in commodity prices.

The risk characteristics mentioned above are reflected in the return-generating process, which makes commodities the ideal time series for testing value-at-risk models: Long tranquil periods alternate with sudden volatility spikes or long periods of high volatility. Some commodity series do not show mean-reverting behaviour, but rather exhibit shifts in volatility. Others switch between positive and negative skewness depending on the time period under investigation. These sudden changes in the return distribution pose a challenge to every VaR model.

Keeping in view the distinct characteristics of commodities, we aim to calculate VaR by using different methodologies and identify the most accurate VaR model for commodities through backtesting techniques.

The paper is structured as follows. In the next section we review the existing literature on VaR methodologies and backtesting. In the third section we discuss the data used in the study. In the fourth section we describe different VaR models and procedures for measuring VaR adequacy used in the study. Fifth section shows the results of analysis and its interpretation. Last section presents the main conclusions of the study.

### **1.1 VaR (Value at Risk)**

“Risk can be defined as a volatility of unexpected outcomes, which can represent the value of assets, equity or earnings” (Jorion, 2007). VaR is a measure of financial risk. Efficient financial risk management allows financial managers to be prepared for potentially abrupt outcomes. In 1970s and 1980s, with a surge in technology; markets expanded by financial derivatives and other assets, created need for a simple risk estimation like VaR. This caused active interest in VaR models. (Angelidis & Benos, 2006), (Berkowitz & O’Brien, 2002), and others employ assumption of normal distribution of returns of the underlying assets in the models as it use is supported by the Central Limit Theorem. However, the suitability of normal distribution for real financial data is still under constant discussion. (Frain, 2008) considers the fit of the actual (historical) data with normal distribution to be unsatisfactory, even misleading. According to Nolan (2003) stable distribution is preferred by some researchers as it fits better tail distribution of actual returns. Nolan shows that some financial sets with heavy tails are well described by stable distributions, or at least much better than by Gaussian models. (Barunik et al., 2010) confirmed that the real data has fatter than normal but lighter than stable tails and the use of stable distribution causes overestimation of extreme returns. (Frain, 2008) confirms this property on models that contain stable distribution. Other distributions are also used by some researchers. For example van den (Goorbergh & Vlaar, 1999) or (Angelidis & Benos, 2006) suggests to model distribution of returns using Student-t-distribution. In a comparison with other volatility models, the EWMA model scored the worst performance in forecasting VaR. It can be seen from

following papers; (Chen et al., 2011), (Abad & Benito, 2013), (Alonso & Arcos, 2006), and (Huang & Lin, 2004). Along this same line, (Giot & Laurent, 2004) compared several volatility models: EWMA, an asymmetric GARCH and RV. The models are estimated with the assumption that returns follow either normal or skewed t-Student distributions. They found that under a normal distribution, the RV model performed best. However, under a skewed t-distribution, the asymmetric GARCH and RV models provided very similar results. These authors reported that the superiority of the models based on RV over the GARCH family is not as obvious when the estimation of the latter assumes the existence of asymmetric and leptokurtic distributions. (Níguez, 2008) compared the ability to forecast VaR of different GARCH family models (GARCH, AGARCH, APARCH, FIGARCH and FIAPARCH, and EWMA) and found that the combination of asymmetric models with fractional integrated models provided the best results. Although this evidence is somewhat ambiguous, the asymmetric GARCH models seem to provide better VaR estimations than the symmetric GARCH models. The strengthened use of VaR models implied need for a model verification by back-testing. Following paragraph reviews the literature on back-testing:

## **1.2 Backtesting**

Despite the wide use and common acceptance of VaR as a risk management tool, the method has frequently been criticized for being incapable to produce reliable risk estimates. When implementing VaR systems, there will always be numerous simplifications and assumptions involved. Moreover, every VaR model attempts to forecast future asset prices using historical market data which does not necessarily reflect the market environment in the future. Thus, VaR models are useful only if they predict future risks accurately. In order to verify that the results acquired from VaR calculations are consistent and reliable, the models should always be backtested with appropriate statistical methods. Backtesting is a procedure where actual profits and losses are compared to projected VaR estimates. Jorion (2001) refers to these tests aptly as ‘reality checks’. If the VaR estimates are not accurate, the models should be re-examined for incorrect assumptions, wrong parameters or inaccurate modelling. Selection of an appropriate VaR methodology to capture market risk accurately is of prime importance with regard to relevance and utility of VaR as a risk measurement and management tool.



A variety of different testing methods have been proposed for backtesting purposes. Basic tests, such as Kupiec's (1995) POF-test, examine the frequency of losses in excess of VaR. This so called failure rate should be in line with the selected confidence level. For instance, if daily VaR estimates are computed at 99% confidence for one year (250 trading days), we would expect on average 2.5 VaR violations, or exceptions, to occur during this period. In the POF-test we would then examine whether the observed amount of exceptions is reasonable compared to the expected amount. This is the technique of backtesting that is used in this study to compare VaR models.

Backtesting is, or at least it should be, an integral part of VaR reporting in today's risk management. Without proper model validation one can never be sure that the VaR system yields accurate risk estimates. The topic is especially important in the current market environment where volatile market prices tend to make investors and more interested in portfolio risk figures as losses accumulate. On the other hand, VaR is known to have severe problems in estimating losses at times of turbulent markets. As a matter of fact, by definition, VaR measures the expected loss only under normal market conditions (e.g. Jorion, 2001). This limitation is one of the major drawbacks of VaR and it makes the backtesting procedures very interesting and challenging, as will be shown later in the study.

## **2. Literature Review**

(Linsmeier & Pearson, 1996) provide introduction to VaR concept and discuss basic methodology. Benefits of logarithmic rate of return over the geometric rate of return are discussed in (Jorion, 2007). Plenty of studies deal with the question of the distribution that approximates empirical returns. First posed by Louis Bachelier in the turn of the 20th century, the Gaussian distribution is used for returns ever since. (Angelidis & Benos, 2006), (Berkowitz & O'Brien, 2002), and others employ normal distribution in the models as its use is supported by the Central Limit Theorem. But the suitability of normal distribution for real financial data is still under constant discussion. (Frain, 2008) considers the fit of the actual (historical) data with normal distribution to be unsatisfactory, even misleading.

Value at Risk by (Jorion, 2007) covers the introduction to the concept of VaR and overall picture of risk and risk measurement. (Christoffersen, 1998), (Christoffersen & Diebold, 2000) and (Christoffersen & Pelletier, 2004) in their work employ conditional coverage whereas (Lopez, 1998) introduces loss function. Basel Committee (Basel Committee, 1996) implemented VaR approach for regulatory purpose and introduced fundamentals for a system of backtesting. From a wide range of works that over overall picture on VaR developments, summarize its properties, individual methods with their strengths and weaknesses, cover the critique towards individual VaR models, and offer alternative approaches to risk management, we draw inspiration mainly from (Jorion, 2007) and (Dowd, 1998). Following works, covering distinctive areas, are source of more detailed information.

(Berkowitz & O'Brien, 2002), and others employ assumption of normal distribution of returns of the underlying assets in the models as it use is supported by the Central Limit Theorem. However, the suitability of normal distribution for real financial data is still under constant discussion. (Frain, 2008) considers the fit of the actual (historical) data with normal distribution to be unsatisfactory, even misleading. According to (Nolan, 2003) stable distribution is preferred by some researchers as it fits better tail distribution of actual returns. Nolan shows that some financial sets with heavy tails are well described by stable distributions, or at least much better than by Gaussian models. (Barunik et al., 2010) confirmed that the real data has fatter than normal but lighter than stable tails and the use of stable distribution causes overestimation of extreme returns. (Frain, 2008) confirms this property on models that contain stable distribution. (Giot & Laurent, 2004) compared several volatility models: EWMA, an asymmetric GARCH and Residual Volatility (RV). The models are estimated with the assumption that returns follow either normal or skewed t-Student distributions. They found that under a normal distribution, the RV model performed best. However, under a skewed t-distribution, the asymmetric GARCH and RV models provided very similar results. These authors reported that the superiority of the models based on RV over the GARCH family is not as obvious when the estimation of the latter assumes the existence of asymmetric and leptokurtic distributions. (Níguez, 2008) compared the ability to forecast VaR of different GARCH family models (GARCH, AGARCH, APARCH, FIGARCH and FIAPARCH, and EWMA) and found that the combination of asymmetric models

with fractional integrated models provided the best results. Although this evidence is somewhat ambiguous, the asymmetric GARCH models seem to provide better VaR estimations than the symmetric GARCH models. The strengthened use of VaR models implied need for a model verification by back-testing.

### **3. Research Gap and Motivation**

For financial institutions, the possibility of sudden losses is unwanted and important to track. Therefore, VaR methodology employs the lower quantiles, which measures only the possibility of an exceptional loss. The strength of VaR measurements lies in the combination of forward looking (ex-ante) risk control, ease of explanation and aggregation of risks. The concept of VaR is quite easy to understand for a wide audience since the overall quantity of potential losses is aggregated into one value incorporating risk from interest rates, equities, commodities and foreign currencies. The pitfall of VaR lies in the necessity to choose a computational method for volatility and a suitable distribution of returns. Choosing an appropriate VaR method is an important and difficult task, and risk managers have coined the term Model Risk to cover the hazards of working with potentially mis-specified models. If a used model underestimates risk, bank's capital buffer might be insufficient to take up the loss or financial regulator may impose penalties. On the other hand, excessive VaR implies an ineffective use of capital. Therefore, it is essential to find a method which accurately specifies the VaR estimate.

The concept of Value at Risk (VaR) has quickly become the standard for measurement of market risk in the financial sector for both financial institutions and financial regulators (Engle & Manganelli, 1999) for its ease of use and clarity of meaning. Systems for internal risk measurement progress in parallel with a regulatory framework. Basel II framework identifies VaR analysis as a preferred tool for assessing exposure of a bank to market risk. As per the Basel report titled “Revisions to the Basel II market risk framework” released in February 2011, following are some key points relevant to our discussion.

- No particular type of model is prescribed. So long as each model used captures all the material risks run by the bank, they will have flexibility in devising the precise nature of their models, but the following minimum standards will apply for the purpose of calculating their capital charge. Individual banks or their supervisory authorities will have discretion to apply stricter standards.
  - a. “Value-at-risk” must be computed on a daily basis.
  - b. In calculating the value-at-risk, a 99th percentile, one-tailed confidence interval is to be used.
  - c. The choice of historical observation period (sample period) for calculating value-at-risk will be constrained to a minimum length of one year. For banks that use a weighting scheme or other methods for the historical observation period, the “effective” observation period must be at least one year.
  - d. Each bank must meet, on a daily basis, a capital requirement expressed as the sum of: The higher of either its previous day’s value-at-risk number or an average of the daily value-at risk measures on each of the preceding sixty business days ( $VaR_{avg}$ ), multiplied by a multiplication factor ( $m_c$ ). The multiplication factors ( $m_c$ ) will be set by individual supervisory authorities on the basis of their assessment of the quality of the bank’s risk management system, subject to an absolute minimum of 3.

Given that banks are permitted to develop differing VaR models, it is necessary for supervisory bodies to be able to assess the relative performance of these alternative models.

VaR is multi-purpose tool that can be used by individual investors, traders, institutional investors, banks etc. to assess the underlying risk of incurring large losses on their asset portfolio. The importance of accurately calculating VaR from a regulatory and compliance point of view is also paramount as discussed above.

#### 4. Objectives

In this study, we use data of nine commodities to calculate VaR and perform backtesting. The results of this study may well be generalized for Indian commodity markets in general. The identification of suitable models to calculate VaR for commodities is useful for traders and investors who deal in commodities in India.

Following are the objectives of the study.

- i. The prime purpose of this paper is to evaluate performance of different approaches and methods of VaR estimation by backtesting. This implies that the objective of this study is to determine the accuracy of VaR models. How can we assess the accuracy and performance of a VaR model? To answer this question, we first need to define what we mean by “accuracy”. By accuracy, we mean: How well does the model measure a particular percentile of the entire profit-and-loss distribution?
- ii. The second aim of this study is to verify whether the selection of similar or contrasting data periods in terms of market conditions for VaR calculation and VaR backtesting lead to a difference in the performance of VaR models. It shall also be interesting to know if this difference leads to preference of different models for VaR estimation according to the extent of congruence between market conditions during VaR calculation and VaR backtesting.

The research hypothesis of the study is that VaR models differ in terms of their accuracy of VaR estimation and that this accuracy depends on the extent of congruence in the market conditions (and thus the returns of underlying asset) during VaR calculation and VaR backtesting periods.

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## 5. Data

The data for this study has been taken from Bloomberg database for nine commodities, all of which are traded on the Multi Commodity Exchange (MCX SX) in India. Daily Closing Prices for all commodities were collected for a period of 9 years starting 25th October 2006 till 23<sup>rd</sup> October 2015. The data points for all nine selected commodities were made consistent by ensuring the inclusion of same dates for the nine year period before dividing it into pre-crisis and post-crisis sets.

VaR calculation and backtesting is performed on nine commodities namely Aluminium, Zinc, Natural Gas, Nickel, Crude Oil, Gold, Silver, Sugar and Copper. This diversified set of commodities spread across categories of agriculture, precious metals, energy etc. is taken as a representative of commodities as an asset class in India.

Many theoretical financial models assume the financial return series to be normally distributed. This is a strong assumption because empirically, financial return series across assets have been found to exhibit negative skewness and excess kurtosis. Therefore it is pertinent to discuss the descriptive statistics of all nine commodities under study.

For the first part of analysis i.e. Pre-Crisis Evaluation, we calculate VaR over a 20 month period i.e. 25th October 2006 till 30th June 2008. The mean daily return ranges from -0.19% (Zinc) to 0.19% (Crude Oil) across all nine commodities for the 20 month period. The standard deviation ranges from 1.1% (Gold) to 3% (Natural Gas) across all 9 commodities for the 20 month period. The skewness ranges from -0.62 (Silver) to 0.23 (Sugar) across all nine commodities for the 20 month period. 8 commodities have negatively skewed returns. Broadly, the returns seem to be symmetric. The Kurtosis ranges from 3.02 (Zinc) to 54.06 (Sugar) across all nine commodities for the 20 month period. All kurtosis values are greater than 3 and indicate a leptokurtic distribution for return series of all commodities. A leptokurtic distribution has fatter tails as compared to a normal distribution. This means that there is a higher possibility of generating extreme returns as compared to a normal distribution. Fat tails in our schemes are evident of the fact that there is higher possibility of generating extreme daily returns (negative or positive) in comparison to a normal distribution. The null hypothesis of normal distribution of returns is

rejected for 7 commodities as Jarque-Bera test's p-value is close to 0 for these commodities. However, Zinc and Nickel have normally distributed returns.

For the second part of analysis i.e. Post-Crisis Evaluation, we calculate VaR over a 4 year period i.e. 1st January 2010 till 31st December 2013. The mean daily return ranges from 0% (Sugar) to 0% (Gold) across all 9 commodities for the 4 year period. The standard deviation ranges from 1.02% (Gold) to 2.6% (Natural Gas) across all 9 commodities for the 4 year period. The skewness ranges from -1.15 (Silver) to 0.24 (Natural Gas) across all 9 commodities for the 4 year period. All commodities have negatively skewed returns except Crude Oil and Natural Gas. The Kurtosis ranges from 4.08 (Aluminium) to 49.8 (Sugar) across all 9 commodities for the 4 year period. The null hypothesis of normal distribution of returns is rejected for all 9 commodities as Jarque-Bera test's p-value is close to 0 for all commodities.

In order to run GARCH (1,1) volatility estimation model, stationarity of parameters has been checked for all nine commodities for pre-crisis and post-crisis analysis. The null hypothesis of unit root (i.e. non-stationarity) in the Augmented Dickey Fuller test is rejected for all nine commodities for pre-crisis and post crisis VaR calculations as the p value is less than 5% level of significance. Hence all nine commodities have a stationary return distribution.

The details of descriptive statistics for pre-crisis and post-crisis VaR calculations are presented in Table 3 and Table 4 of Annexure 1 respectively.

## **6. Methodology**

VaR is calculated using historical data. This implies that if VaR is calculated in a relatively tranquil period with positive returns on an average, it shall produce an estimate which is not suitable for a crisis situation having large negative returns and vice-versa. This indicates that VaR is heavily dependent on historical data and assumes that the returns in near future shall be consistent with the returns in the past. Keeping this in view, we decided to divide our 9 years of daily return data into two parts: Pre-Crisis Evaluation and Post-Crisis Evaluation. Our total data starting in October 2006 and ending in October 2015 has a phase of severely distressed returns

during the period of financial crisis of 2008-2009. This global meltdown owing to the sub-prime mortgage crisis in the US had its fair share of impact on India too.

According to (NBER, 2010), the US business cycle reached its peak in December 2007 and trough in June 2009. To mark the crisis period accordingly, we have taken a period of 18 months starting 1st July 2008 till 31st December 2009 to represent this phase of financial distress. Although the financial crisis had started showing its symptoms in early 2008, we decided to mark its beginning in July 2008 as the crisis had a delayed and relatively weaker effect on the Indian markets. Similarly, although the trough might have been reached in June 2009, since the economy takes time to recover and equity returns stabilize gradually, we decided to add another 6 months and stretch the period of distress to 31st December 2009 making it a 18 month long phase.

VaR backtesting has been applied by adopting the intuitive out-of-sample approach. This means that VaR is first calculated over a period of time and then backtesting is applied ahead of this time to test whether VaR estimates are able to stand the test of time by predicting frequency of violations accurately or not. This out-of-sample approach is appropriate to compare VaR methods because it examines the relevance of a VaR estimate as a forward looking risk measure. It helps us identify which methods are able to produce future-oriented estimates on the basis of historical data. If a method sustains the out-of-sample backtesting approach and produces lesser number of violations than the significance level, it is an appropriate way of calculating VaR.

For the first part of analysis i.e. Pre-Crisis Evaluation, we calculate VaR over a 20 month period i.e. 25th October 2006 till 30th June 2008 and backtest the estimates over a 18 month period of financial crisis i.e. 1st July 2008 to 31st December 2009. This implies that we try to find out which methods produce accurate estimates of VaR (on the basis of historical data during period of relatively stable returns) which are able to withstand backtesting during the turbulent phase of financial crisis.

For the second part of analysis i.e. Post-Crisis Evaluation, we calculate VaR over a 4 year period i.e. 1st January 2010 till 31st December 2013 and backtest the estimates over a 22 month period i.e. 1st January 2014 to 25th October 2015. This implies that we try to find out if there is a



difference between quality of VaR estimates produced by different methods if the VaR calculation period and backtesting period are largely similar in terms of economic conditions.

After dividing the data into relevant periods for a two-fold analysis, we calculate VaR by using following methods.

### 6.1 Parametric VaR Methods

- a. **Normal VaR:** VaR is calculated with the assumption that returns follow a normal probability distribution. The historical volatility is calculated as standard deviation of the return series and plugged in the formula for VaR. The one tailed standard normal variate corresponding to the chosen confidence level is multiplied by the volatility to calculate the extent of deviation from the mean to arrive at VaR figure.

$$VaR(\alpha) = F^{-1}(\alpha) = \mu_t + \sigma_t(\alpha)$$

Where,  $\mu$  represents mean,  $\sigma$  represents historical standard deviation,  $\alpha$  denotes confidence level and  $t$  denotes time period considered.

- b. **GARCH (1,1) VaR:** The only difference between Normal VaR and GARCH (1,1) VaR is the difference in calculation of volatility to be inserted in the VaR calculation. This methodology produces a forward looking estimate of volatility and hence can predict VaR more accurately if there is volatility clustering in the return series. In this method,  $\sigma$  is calculated using Exponential Generalised Auto-Regressive Conditional Heteroskedasticity (GARCH (1,1)). Exponential GARCH is estimated using following equations:

$$R_t = \mu_t + \varepsilon_t$$

$$\sigma_t^2 = \omega + \sum_{j=1}^q \alpha_j [\varepsilon_{t-j}^2] + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

Where,  $R_t$  represents return,  $\mu_t$  represents mean return and  $\varepsilon_t$  is error term in mean equation;  $\sigma_t$  is volatility, coefficient  $\omega$  is constant,  $\alpha_j$  represents ARCH term and  $\beta_j$  represents GARCH term.

## 6.2 Non-parametric VaR Method

### a. Historical Simulation VaR

In this, VaR is estimated, without taking into consideration any assumption about underlying distribution. It is based on the assumption that history is repeating itself. It is the simplest method of estimating VaR. The justification for this methodology is that if returns are stationary then empirical distribution is a consistent estimator of the unobserved future distribution function. This method is defined as follows.

Consider a sample of past  $\omega$  returns. The historical simulation VaR at  $\alpha$  level of significance for period  $t+1$  is given by:

$$VaR_{\alpha} = -Q_{1-\alpha}(r_t, r_{t-1}, \dots, r_{t-\omega+1})$$

Where,  $r_t$  is return of the asset under consideration at time  $t$  and  $Q$  is the relevant quantile function at  $\alpha$  level of significance.

### b. Historical Simulation using Bootstrapping

Bootstrap historical simulation approach is an extension of traditional historical simulation. It is a simple and intuitive estimation procedure. The bootstrap technique draws a sample from the data set, records the VaR from that particular sample and “returns” the data. This procedure is repeated over and over and records multiple sample VaRs. Since the data is always returned to the data set, this procedure is like sampling with replacement. The best VaR estimate from the full data set is the average of all sample VaRs.

**6.3 Semi-Parametric Methods:** The process combines the traditional simulation model with conditional volatility models like GARCH (1,1) which makes it attractive in dealing with volatility features. In this category, following methodologies are used in this study.

### a. Volatility-Weighted Historical Simulation

This method was proposed by (Hull & White, 1998). It combines the benefits of historical simulation with volatility updating. The basic premise of this approach is to update return

information with recent changes in volatility. According to this approach, VaR ( $\alpha$ ) is the  $\alpha$  quantile of the distribution of the volatility adjusted return. In this methodology all past returns are scaled by volatility adjustment factor. Volatility adjusted returns are as follows.

$$R_{t,i} = r_{t,i} * \frac{\sigma_{T,i}}{\sigma_{t,i}}$$

Where,  $r_{t,i}$  is actual return for asset  $i$  on day  $t$ ;  $\sigma_{T,i}$  is current forecast of volatility for asset  $i$ ;  $\sigma_{t,i}$  is volatility forecast for asset  $i$  on day  $t$  (made at the end of day  $t-1$ ).

#### 6.4 Backtesting of VaR Models

The most common backtesting method for a VaR model is to count the number of VaR exceptions, i.e. days (or holding periods of other length) when portfolio losses exceed VaR estimates. This method is known as the Kupiec Test as it was suggested by Paul Kupiec in 1995. If the number of exceptions is less than the selected confidence level would indicate, the system overestimates risk. On the contrary, too many exceptions signal underestimation of risk. Naturally, it is rarely the case that we observe the exact amount of exceptions suggested by the confidence level. It therefore comes down to statistical analysis to study whether the number of exceptions is reasonable or not, i.e. will the model be accepted or rejected.

Denoting the number of exceptions as  $x$  and the total number of observations as  $T$ , we may define the failure rate as  $x/T$ . In an ideal situation, this rate would reflect the selected confidence level. For instance, if a confidence level of 99 % is used, we have a null hypothesis that the frequency of tail losses is equal to  $p = (1 - c) = 1 - .99 = 1\%$ . Assuming that the model is accurate, the observed failure rate  $x/T$  should act as an unbiased measure of  $p$ , and thus converge to 1% as sample size is increased. (Jorion, 2001)

Each trading outcome either produces a VaR violation exception or not. This sequence of ‘successes and failures’ is commonly known as Bernoulli trial. The number of exceptions  $x$  follows a binomial probability distribution:

$$f(x) = p^x(1 - p)^{T-x}$$

As the number of observations increase, the binomial distribution can be approximated with a normal distribution:

$$z = \frac{x - pT}{\sqrt{p(1-p)T}} \approx N(0,1)$$

where  $pT$  is the expected number of exceptions and  $p(1-p)T$  is the variance of exceptions. (Jorion, 2001)

By utilizing this binomial distribution we can examine the accuracy of the VaR model. However, when conducting a statistical backtest that either accepts or rejects a null hypothesis (of the model being ‘good’), there is a trade-off between two types of errors. Type 1 error refers to the possibility of rejecting a correct model and type 2 error to the possibility of not rejecting an incorrect model. A statistically powerful test would efficiently minimize both of these probabilities. (Jorion, 2001)

Hence, the only information required to implement a POF-test is the number of observations ( $T$ ), number of exceptions ( $x$ ) and the confidence level ( $c$ ). (Dowd, 2006)

## **7. Analysis and Interpretation**

We have divided our analysis into separate two heads viz. Pre Crisis Evaluation and Post Crisis Evaluation. We shall start by studying the results of Pre Crisis Evaluation. The key phenomenon to keep in consideration is that for Pre Crisis Evaluation, we calculate VaR over a 20 month period i.e. 25th October 2006 till 30th June 2008 and backtest the estimates over a 18 month period of financial crisis i.e. 1st July 2008 to 31st December 2009. The VaR calculation and backtesting periods are contrasting in terms of their respective market conditions. Following is a table showing results of the Pre-Crisis Evaluation.

**Table 1: Pre-Crisis Evaluation of VaR Models**

| <b>Category of VaR Method</b>                                        | <b>Non Parametric</b> |                                    | <b>Parametric</b> |                        | <b>Semi Parametric</b> |
|----------------------------------------------------------------------|-----------------------|------------------------------------|-------------------|------------------------|------------------------|
| <b>Null Hypothesis: <math>x/T = \text{Significance Level}</math></b> | <b>Historical VaR</b> | <b>Bootstrapped Historical VaR</b> | <b>Normal VaR</b> | <b>GARCH (1,1) VaR</b> | <b>VWHS VaR</b>        |
| Hypotheses 'Not Rejected' out of 9                                   | 3                     | 2                                  | 2                 | 3                      | 9                      |
| Success Ratio                                                        | 33.33%                | 22.22%                             | 22.22%            | 33.33%                 | 100%                   |

The null hypothesis being tested in the Kupiec test is that proportion of failure i.e.  $x/T$  ( $x$  is the number of violations and  $T$  is the total number of observations) is equal to  $\alpha$  i.e. level of significance. If this null hypothesis is not rejected, the VaR method passes the Kupiec Test. This is because the total number of violations observed in the testing period is not statistically different from the level of significance. The level of significance has been fixed at 5% for all hypotheses tests conducted in this study. For each VaR method, the success ratio has been defined as ratio of number of commodities for which the null hypotheses is not rejected and total number of commodities i.e. nine.

As can be observed from the above table, out of the total 9 commodities for which VaR has been calculated and backtested, Volatility Weighted Historical Simulation (VWHS) VaR has been able to accurately predict VaR figures for all 9 commodities i.e. the success ratio of VWHS VaR is 100%. This means that there is no difference between the proportion of failure observed in the backtesting period and the significance level of 5%. This shows that standardizing the daily log returns by GARCH volatility estimates produces a return series in such a way that it gives accurate VaR estimates calculated as the relevant percentile of the distribution.

Parametric VaR models viz. Normal VaR and GARCH (1,1) VaR have performed poorly in the backtest with a success ratio of 22.22% and 33.33% respectively. This is mainly due to the use of standard normal variate to determine VaR quantile whereas the return distribution is far from normal.

Non-Parametric methods viz. Historical VaR and Bootstrapped historical VaR have also performed poorly in the backtesting analysis with a success ratio of 22.22% each.

The detailed table consisting of VaR estimates and result of hypothesis test for each pair of commodity and VaR method pertaining to pre-crisis evaluation is provided in Annexure II.

We now move on to the analysis of results of Post-Crisis Evaluation. In this case, we calculate VaR over a 4 year period i.e. 1st January 2010 till 31st December 2013 and backtest the estimates over a 22 month period i.e. 1st January 2014 to 25th October 2015.

The period of VaR calculation and VaR backtesting are broadly similar in the sense that none of the periods had experienced a situation of distressed returns. It is interesting to know whether different VaR methods perform similarly if there is no significant difference between the VaR calculation and VaR backtesting period. Following is a table showing results of the Post-Crisis Evaluation.

**Table 2: Post-Crisis Evaluation of VaR Models**

| <b>Category of VaR Method</b>                                        | <b>Non Parametric</b> |                                    | <b>Parametric</b> |                        | <b>Semi Parametric</b> |
|----------------------------------------------------------------------|-----------------------|------------------------------------|-------------------|------------------------|------------------------|
| <b>Null Hypothesis: <math>x/T = \text{Significance Level}</math></b> | <b>Historical VaR</b> | <b>Bootstrapped Historical VaR</b> | <b>Normal VaR</b> | <b>GARCH (1,1) VaR</b> | <b>VWHS VaR</b>        |
| Hypotheses 'Not Rejected' out of 9                                   | 7                     | 7                                  | 6                 | 3                      | 9                      |
| Success Ratio                                                        | 77.78%                | 77.78%                             | 66.67%            | 33.34%                 | 100%                   |

We observe that for post crisis evaluation, VWHS VaR has again produced a success ratio of 100%. In fact, Historical VaR and Bootstrapped Historical VaR also give highly accurate results with a success ratio of 77.78% each. Normal VaR exhibits a success ratio of 66.67%.

All methods except GARCH (1,1) VaR perform broadly well in this scenario where the data used for VaR calculation is quite similar in its behaviour to the data used for VaR backtesting. It is because of this similarity in results that our first analysis of pre-crisis evaluation becomes even more important to differentiate among VaR methods

However, VWHS VaR has emerged as a clear winner in both parts of the analysis. It has accurately predicted VaR estimates in pre-crisis and post-crisis evaluation delivering 100% success ratio each time. Hence, we conclude that VWHS VaR is the best methodology out of the five different methods compared in this study for measuring VaR of a portfolio of commodities in India.

The detailed table consisting of VaR estimates and result of hypothesis test for each pair of commodity and VaR method pertaining to post-crisis evaluation is provided in Annexure III.

## **8. Conclusion and Implications**

Value at Risk is widely used measure of risk today because of its simplicity and flexibility. The variants of VaR are evident its suitability and universal appeal as a standard measure of risk. It is a measure that needs to be reported by commercial banks as a regulatory compliance according to the Basel norms. But the flexibility of VaR is also one of its limitations. Different approaches to calculate VaR are based on different assumptions. These assumptions relate to the underlying distribution of returns and volatility measurement. It is thus important to identify the most appropriate method of calculating VaR for different assets and markets. In this study we tried to identify the most suitable VaR methodology for a portfolio of commodities traded on Multi Commodity Exchange of India. To get a broad representation of commodities, we performed Kupiec's test of backtesting on nine commodities over a two-fold period separated by the financial crisis of 2008-09. The VaR methodologies considered were Historical Simulation VaR, Bootstrapped Historical VaR, Normal VaR, GARCH (1,1) VaR and Volatility Weighted Historical Simulation (VWHS) VaR. In our study, VWHS VaR has outperformed other methods in both parts of the analysis exhibiting a success ratio of 100% each time. Non Parametric methods performed poorly in Pre Crisis analysis but relatively well in Post Crisis period. However, parametric methods have performed poorly in accurately estimating VaR in both parts of the analysis. Hence, we conclude that VWHS VaR is the best methodology out of the five different methods compared in this study for measuring VaR of standalone investments in a commodity or a portfolio of commodities in India. We also conclude that the selection of similar

or contrasting data periods in terms of market conditions for VaR calculation and VaR backtesting affects the performance of VaR methods in general. However, it did not lead to a change in supremacy of VWHS VaR across the two periods of analysis in our study.



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## Annexure I

**Table 3: Descriptive Statistics: Pre-Crisis Analysis**

|                     | Aluminium | Zinc     | Nat Gas  | Nickel   | Crude Oil | Gold     | Silver   | Sugar    | Copper   |
|---------------------|-----------|----------|----------|----------|-----------|----------|----------|----------|----------|
| <b>Mean</b>         | 0.000162  | -0.00193 | 0.001065 | -0.00115 | 0.001989  | 0.000921 | 0.000715 | -0.0004  | 0.000163 |
| <b>Median</b>       | 0         | -0.00148 | 0.001244 | 0        | 0.001722  | 0.001146 | 0.002043 | -0.00068 | 0        |
| <b>Maximum</b>      | 0.059189  | 0.069733 | 0.150331 | 0.084943 | 3.801019  | 0.036604 | 0.063067 | 0.149812 | 0.060615 |
| <b>Minimum</b>      | -0.05428  | -0.07702 | -0.14885 | -0.09421 | -3.7991   | -0.05336 | -0.06549 | -0.15059 | -0.07817 |
| <b>Std. Dev.</b>    | 0.015238  | 0.025049 | 0.030125 | 0.027153 | 0.319216  | 0.011137 | 0.015465 | 0.017508 | 0.02048  |
| <b>Skewness</b>     | -0.04656  | -0.09857 | -0.12903 | -0.10444 | -0.01016  | -0.23718 | -0.62743 | 0.238854 | -0.26626 |
| <b>Kurtosis</b>     | 4.118058  | 3.020456 | 6.064447 | 3.425164 | 117.6244  | 4.709059 | 6.345122 | 54.06256 | 4.258258 |
| <b>Jarque-Bera</b>  | 22.13262  | 0.690677 | 166.2931 | 3.94561  | 231023.2  | 55.31555 | 224.4425 | 45850.53 | 32.82426 |
| <b>p-value</b>      | 1.56E-05  | 0.707981 | 0        | 0.139066 | 0         | 9.74E-13 | 0        | 0        | 7.45E-08 |
| <b>Observations</b> | 422       | 422      | 422      | 422      | 422       | 422      | 422      | 422      | 422      |

**Table 4: Descriptive Statistics: Post-Crisis Analysis**

|                     | Aluminium | Zinc     | Nat Gas  | Nickel   | Crude Oil | Gold     | Silver   | Sugar    | Copper   |
|---------------------|-----------|----------|----------|----------|-----------|----------|----------|----------|----------|
| <b>Mean</b>         | 5.90E-05  | 7.38E-05 | 2.87E-05 | 2.34E-06 | 0.000512  | 0.000561 | 0.000479 | -0.00027 | 0.000297 |
| <b>Median</b>       | 0         | 0        | 0        | 0.000213 | 0         | 0.000281 | 0.000209 | -0.00067 | 0        |
| <b>Maximum</b>      | 0.052432  | 0.06908  | 0.131891 | 0.055417 | 0.10194   | 0.043873 | 0.079697 | 0.119553 | 0.069446 |
| <b>Minimum</b>      | -0.06417  | -0.07102 | -0.08894 | -0.12118 | -0.09084  | -0.08793 | -0.16449 | -0.13803 | -0.08647 |
| <b>Std. Dev.</b>    | 0.013161  | 0.016074 | 0.026913 | 0.017081 | 0.018063  | 0.010265 | 0.019413 | 0.010886 | 0.016235 |
| <b>Skewness</b>     | -0.23743  | -0.30612 | 0.244747 | -0.5874  | 0.140535  | -1.08827 | -1.15023 | -0.33158 | -0.12134 |
| <b>Kurtosis</b>     | 4.089999  | 4.158842 | 4.303992 | 6.021621 | 6.06696   | 12.39169 | 13.87831 | 49.80391 | 4.98099  |
| <b>Jarque-Bera</b>  | 58.36932  | 70.9292  | 80.1058  | 433.9905 | 391.6616  | 3837.693 | 5104.879 | 90471.95 | 164.4736 |
| <b>p-value</b>      | 2.11E-13  | 4.44E-16 | 0        | 0        | 0         | 0        | 0        | 0        | 0        |
| <b>Observations</b> | 991       | 991      | 991      | 991      | 991       | 991      | 991      | 991      | 991      |

## Annexure II

### Results of VaR Backtesting: Pre-Crisis Analysis

|                                                          |           | <b>Historical<br/>VaR</b> | <b>Normal<br/>VaR</b> | <b>Historical<br/>Simulation<br/>VaR using<br/>bootstrap</b> | <b>GARCH<br/>(1,1) VaR</b> | <b>VVHS<br/>VaR</b> |
|----------------------------------------------------------|-----------|---------------------------|-----------------------|--------------------------------------------------------------|----------------------------|---------------------|
| <b>Aluminium</b>                                         | VaR Value | 0.039649206               | 0.03528631            | 0.03961610                                                   | 0.03664894                 | 0.04015181          |
|                                                          | Decision  | Rejected                  | Rejected              | Rejected                                                     | Rejected                   | Accepted            |
| <b>Zinc</b>                                              | VaR Value | 0.061452779               | 0.06020445            | 0.06032271                                                   | 0.05916738                 | 0.05633088          |
|                                                          | Decision  | Rejected                  | Rejected              | Rejected                                                     | Rejected                   | Accepted            |
| <b>Natural Gas</b>                                       | VaR Value | 0.077436654               | 0.06901550            | 0.07823509                                                   | 0.05448986                 | 0.0881792           |
|                                                          | Decision  | Rejected                  | Rejected              | Rejected                                                     | Rejected                   | Accepted            |
| <b>Nickel</b>                                            | VaR Value | 0.06928471                | 0.06431264            | 0.06878288                                                   | 0.06103846                 | 0.06446742          |
|                                                          | Decision  | Rejected                  | Rejected              | Rejected                                                     | Rejected                   | Accepted            |
| <b>Crude Oil</b>                                         | VaR Value | 0.050269268               | 0.74061953            | 0.04964276                                                   | 0.12239652                 | 0.37505467          |
|                                                          | Decision  | Rejected                  | Accepted              | Rejected                                                     | Accepted                   | Accepted            |
| <b>Gold</b>                                              | VaR Value | 0.02825                   | 0.02498710            | 0.02756968                                                   | 0.03644088                 | 0.02616895          |
|                                                          | Decision  | Rejected                  | Rejected              | Rejected                                                     | Accepted                   | Accepted            |
| <b>Silver</b>                                            | VaR Value | 0.053810449               | 0.03526270            | 0.05269958                                                   | 0.03793999                 | 0.04812322          |
|                                                          | Decision  | Accepted                  | Rejected              | Accepted                                                     | Rejected                   | Accepted            |
| <b>Sugar</b>                                             | VaR Value | 0.026233337               | 0.04112591            | 0.04230372                                                   | 0.02460733                 | 0.04883971          |
|                                                          | Decision  | Accepted                  | Accepted              | Accepted                                                     | Accepted                   | Accepted            |
| <b>Copper</b>                                            | VaR Value | 0.064004284               | 0.04748062            | 0.06115966                                                   | 0.04188975                 | 0.08056324          |
|                                                          | Decision  | Accepted                  | Rejected              | Rejected                                                     | Rejected                   | Accepted            |
| <b>Total number of<br/>Hypotheses 'Not<br/>Rejected'</b> |           | <b>3</b>                  | <b>2</b>              | <b>2</b>                                                     | <b>3</b>                   | <b>9</b>            |

### Annexure III

#### Results of VaR Backtesting: Post-Crisis Analysis

|                                                  |           | Historical VaR | Normal VaR | Historical Simulation VaR using bootstrap | GARCH (1,1) VaR | VVHS VaR |
|--------------------------------------------------|-----------|----------------|------------|-------------------------------------------|-----------------|----------|
| <b>Aluminium</b>                                 | VaR Value | 0.035303       | 0.030559   | 0.034199                                  | 0.025368        | 0.037855 |
|                                                  | Decision  | Accepted       | Accepted   | Accepted                                  | Accepted        | Accepted |
| <b>Zinc</b>                                      | VaR Value | 0.04353        | 0.037319   | 0.043557                                  | 0.025318        | 0.048607 |
|                                                  | Decision  | Accepted       | Accepted   | Accepted                                  | Rejected        | Accepted |
| <b>Natural Gas</b>                               | VaR Value | 0.06732        | 0.06258    | 0.0651                                    | 0.045157        | 0.062674 |
|                                                  | Decision  | Rejected       | Rejected   | Rejected                                  | Rejected        | Accepted |
| <b>Nickel</b>                                    | VaR Value | 0.04398        | 0.039734   | 0.044098                                  | 0.02665         | 0.039712 |
|                                                  | Decision  | Accepted       | Rejected   | Accepted                                  | Rejected        | Accepted |
| <b>Crude Oil</b>                                 | VaR Value | 0.050034       | 0.041508   | 0.048497                                  | 0.028048        | 0.065038 |
|                                                  | Decision  | Rejected       | Rejected   | Rejected                                  | Rejected        | Accepted |
| <b>Gold</b>                                      | VaR Value | 0.031203       | 0.023318   | 0.030659                                  | 0.017384        | 0.033703 |
|                                                  | Decision  | Accepted       | Accepted   | Accepted                                  | Accepted        | Accepted |
| <b>Silver</b>                                    | VaR Value | 0.058474       | 0.044681   | 0.058811                                  | 0.038073        | 0.070489 |
|                                                  | Decision  | Accepted       | Accepted   | Accepted                                  | Accepted        | Accepted |
| <b>Sugar</b>                                     | VaR Value | 0.027175       | 0.025592   | 0.029269                                  | 0.006752        | 0.027743 |
|                                                  | Decision  | Accepted       | Accepted   | Accepted                                  | Rejected        | Accepted |
| <b>Copper</b>                                    | VaR Value | 0.044321       | 0.037473   | 0.043287                                  | 0.022661        | 0.056463 |
|                                                  | Decision  | Accepted       | Accepted   | Accepted                                  | Rejected        | Accepted |
| <b>Total number of Hypotheses 'Not Rejected'</b> |           | <b>7</b>       | <b>6</b>   | <b>7</b>                                  | <b>3</b>        | <b>9</b> |